

José Miguel Ramírez Gutiérrez

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EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM) Querétaro, Mexico
B.S. in Robotics and Digital Systems Engineering (GPA: 3.7/4.0) Aug. 2024 – Expected 2028
• Socioeconomic Scholarship

EXPERIENCE

Automation Developer (PEPS Validation) June 2026
Valeo (Contractor) Querétaro, Mexico
• Automated manual Key Fob testing procedures by developing a complete Positioning System software architecture, allowing engineers to save up to 8 hours daily in testing.
• Built an autonomous workspace calibration system using 4 HC-SR04 ultrasonic sensors with a halving-approach algorithm, achieving ± 1 cm positioning accuracy over a 390×1300
• Programmed a responsive Python/Tkinter GUI for real-time control and automated movement sequencing, streamlining hardware communication and execution of validation scripts.
• Developed robust CoreXY kinematics firmware in C++ for non-blocking stepper motor control, enabling precise and continuous physical execution of testing coordinates.

Cloud Integration & IT Specialist 2026 – Present
binarysails Querétaro, Mexico
• Collaborated to verify and standardize Microsoft OneDrive functionality, remote sharing, and synchronization across multiple external computers.
• Professionalized the delivery of project documentation by standardizing the organization of electronic and firmware files within shared cloud drives.

PROJECTS

Custom Linux Single Board Computer (SBC) | KiCad, Allwinner V3s, Hardware Design July 2026
• Engineered a custom 4-layer PCB around the Allwinner V3s (ARM Cortex-A7) processor, integrating an EA3036CQB PMIC, Ethernet PHY, and onboard RTC.
• Implemented differential pair routing with controlled impedance to guarantee signal integrity for High-Speed USB 2.0 (480 Mbps) communication.
• Optimized board layout and component placement for automated SMT manufacturing (PCBA), resolving centroid rotation discrepancies and footprint constraints.

ESP32 AI Voice Assistant | MicroPython, KiCad May 2026
• Designed a custom PCB in KiCad integrating an ESP32, INMP441 I2S microphone, PAM8403 amplifier, and SH1106 OLED display; generated and ordered production-ready Gerber files.
• Achieved rapid response times by integrating Groq API (Whisper + LLaMA 3.1) for audio transcription and LLM processing via a Flask backend deployed on Render.
• Implemented wake-word detection with VAD-based audio capture.

DetectaGas – IoT Gas Leak Prevention System | ESP32, Rust, React, Axum December 2025
• Engineered an IoT device using an ESP32 and MQ-5 sensor to rapidly reduce response time to hazardous LP gas leaks.
• Built a concurrent Rust backend with Axum and a React web interface, processing continuous telemetry via an MQTT broker.

Flight Monitoring & Data Analysis | Python January 2026
• Processed and cleaned structured datasets of commercial flights using Python, improving overall trend visualization accuracy.
• Developed modular scripts for scalability, drastically reducing data preprocessing time for real-world analytical tasks.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, C#, JavaScript, MATLAB, VHDL
Embedded & Hardware: ESP32, Arduino, KiCad (PCB Design)
Software & Technologies: MicroPython, Arduino IDE, Axum, MQTT, MySQL, Git, React, Flask
Spoken Languages: Spanish (Native), English (Business Fluent)